

3.3.2 SYNCHROS

fig. Synchros is the abbreviation of the word 'synchronous'. It is the name given to a variety of rotary, electromechanical devices or electromagnetic transducer used for sensing angular position of shafts. Synchros make possible the accurate transmission and reproduction of a remote location of any data or information. With proper cables, the distance between the transmitter and receiver can be as far as 5km. Pointer or indicator on the receiver shaft will smoothly follow the motion of the transmitter shaft. Synchros play a very important role in the operation of Navy equipment. Synchros are found in just every weapon system, communication satellite antennas, underwater detection system, and navigation system used in the Navy.

A synchro system is formed by interconnection of two or more synchros. There are two general classification of synchros system – 'Torque system' and 'control system'. Torque synchros systems are torque synchros and control-synchros systems used to control synchros. The load dictates the type of synchros system and thus the type of synchros. Torque synchros system are classified 'torque' because they are mainly concerned with the torque or turning force required to move light loads such as dials or similar indicators. The positioning of these devices require a relatively low amount of torque. Control – synchros are used in systems that are designed to move heavy load such as gun directors radar antennas and missile launcher. Hence, the name torque transmitter (TT) Torque receiver (TR) control transmitter (CT) and control receiver (CR) respectively.

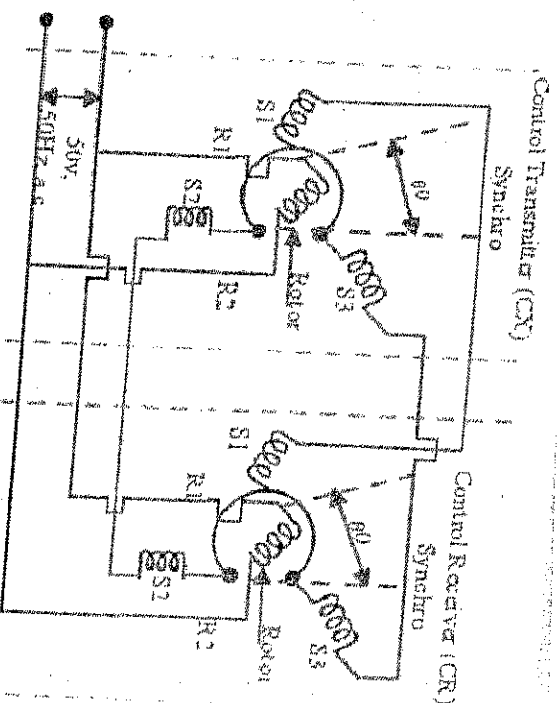


Fig. 3.3 Control Transmitter (CT) to control receiver (CR) synchro system.

4.0 Process control systems

4.1 Process

Process refers to the methods of changing or refining the raw-materials to create end products. The raw materials, which either pass through or remain in a liquid gaseous, or slurry (a mix of solid and liquids) state during the process, are transferred, measured, mixed, heated or cooled, filtered, stored, or handled in some other way to produce the end product. In this context, a process is a combination of materials and equipment that produces a desirable results through changes in energy, physical properties or chemical properties. Process industries include the chemical industry, the oil and gas industry, the food and beverage industry, the pharmaceutical industry, the water treatment industry, and the power industry, petroleum refinery, fertilizer plant, home heating system, electric power plant, etc.

4.2 Process control

Process control refers to the methods that are used to regulate variables when manufacturing a product. For example, factors such as the proportion of one

ingredient to another, the temperature of the materials, how well the ingredients are mixed and the pressure under which the materials are held can significantly impact the quality of an end product. Manufacturers control the production process for three reasons, namely to;

- Reduce variability
- Increase efficiency
- Ensure safety.

Common process variables

The most common controlled variables in a process are;

1. Temperature
2. Pressure
3. Flow rate
4. Liquid level
5. Density
6. Viscosity
7. Composition
8. Colour
9. Conductivity
10. PH
11. hardness
12. thickness, etc

Process control terms

Process control has its peculiar terms some of which are discussed hereunder,

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4.1 Set point

The set point is a value for a process variable to be maintained constant. For example, if a process temperature needs to be kept at 100°C , then the set point is 100°C . A temperature sensor can be used to help maintain the temperature at set point. The sensor inserted into the process, and a controller compares the temperature reading from the sensor to the set point. If the temperature reading is different, the controller signals the heater to take corrective action so as to bring the temperature back to the set point.

4.4.2 Deviation (offset)

The deviation or offset is the difference between the process output and the setpoint. In the temperature control system example, if the control system holds the temperature at 102°C consistently, even though the set point is 100°C , then an offset of 2°C exists.

4.4.3 Transport delay (distance-velocity lag)

A transport delay occurs whenever mass or fluid is transported from one point to another. It is the time required for the mass or fluid to travel from the input location to the output location. Transport delay is also called transport time, distance-velocity lag or dead-time delay. Control process becomes increasingly more difficult as dead time increases.

The dead-time delay of pipe line is equal to the distance travelled (D) divided by the average velocity (V) of the fluid. That is;

$$t_d = \frac{D}{V}, \text{ but } V = \frac{Q}{a} \text{ m/s, also } a = \frac{\pi d^2}{4}$$

$$\text{Where } Q = \text{Flow rate}$$

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4.4.4 Example

Liquid flows in a pipe that is 200m long and has a diameter of 6cm. The flow rate is 0.0113m³/s. Determine the transport-delay.

Solution

Given that,

$$\text{Flow-rate } Q = 0.0113 \text{ m}^3/\text{s}$$

$$\text{Pipe distance } D = 200 \text{ m}$$

$$\text{Pipe Diameter } d = 6 \text{ cm} = 0.06 \text{ m}$$

$$\text{Transport delay } t_d = ?$$

$$\text{Average velocity of liquid } v = \frac{4Q}{\pi d^2} \text{ m/s}$$

$$= \frac{4 \times 0.0113}{\pi \times (0.06)^2}$$

$$= \frac{0.0452}{0.0036 \pi^2} \text{ m/s}$$

$$= \frac{0.0452}{0.0113097}$$

$$= 3.996 \text{ m/s}$$

$$\text{Hence, transport delay } t_d = \frac{D}{v} = \frac{200}{3.966}$$

$$= 50.04 \text{ seconds}$$

4.5 Automatic temperature control process

The arrangement for automatic temperature control system is shown in Fig. 4.5.1 consists of an electric furnace whose temperature is detected by the thermistor. The reference temperature which corresponds to a particular voltage is set by the wiper of the potentiometer and fed to the “+” input of the comparator. The temperature level detected by the thermistor is fed through a temperature to voltage converter to the “-” input of the comparator. This feedback process temperature is compared with the reference or set temperature and the difference or error signal is supplied to the controller which controls the level of power applied to the heater of the furnace. With this method temperature of the furnace is automatically controlled.

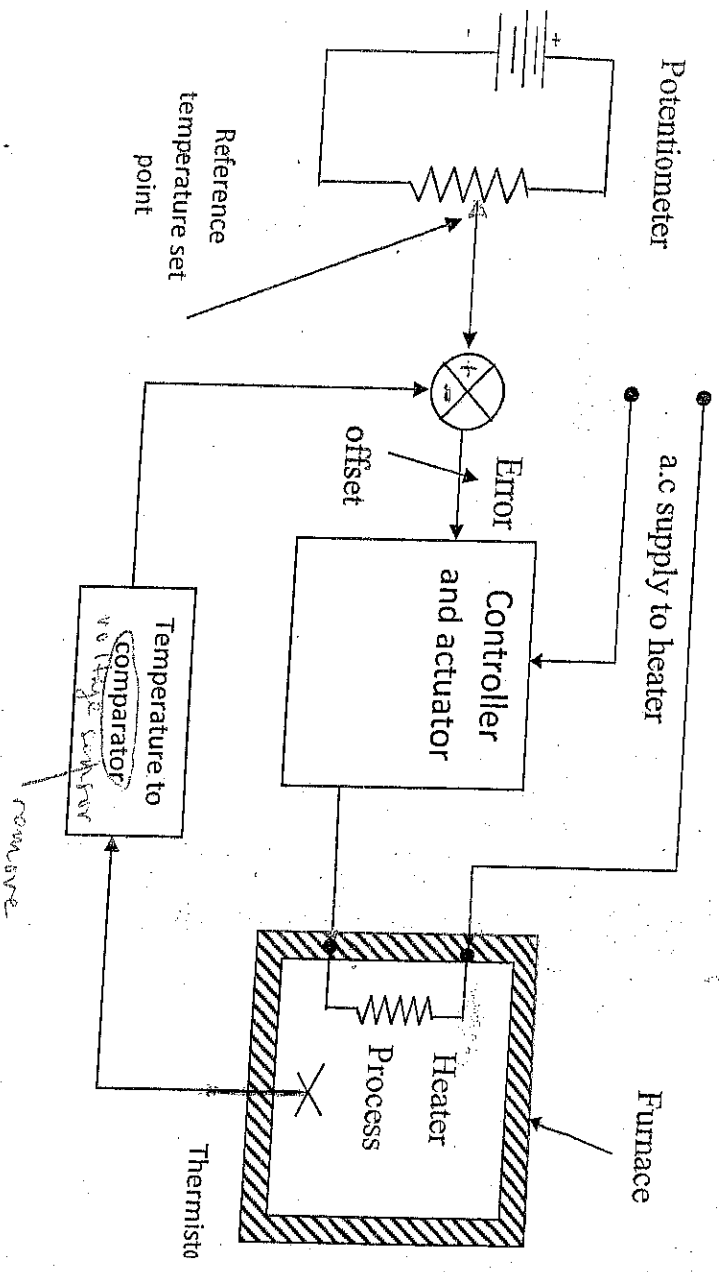


Fig. 4.5 Automatic temperature control system

Automatic liquid-level control process

Fig. 4.5
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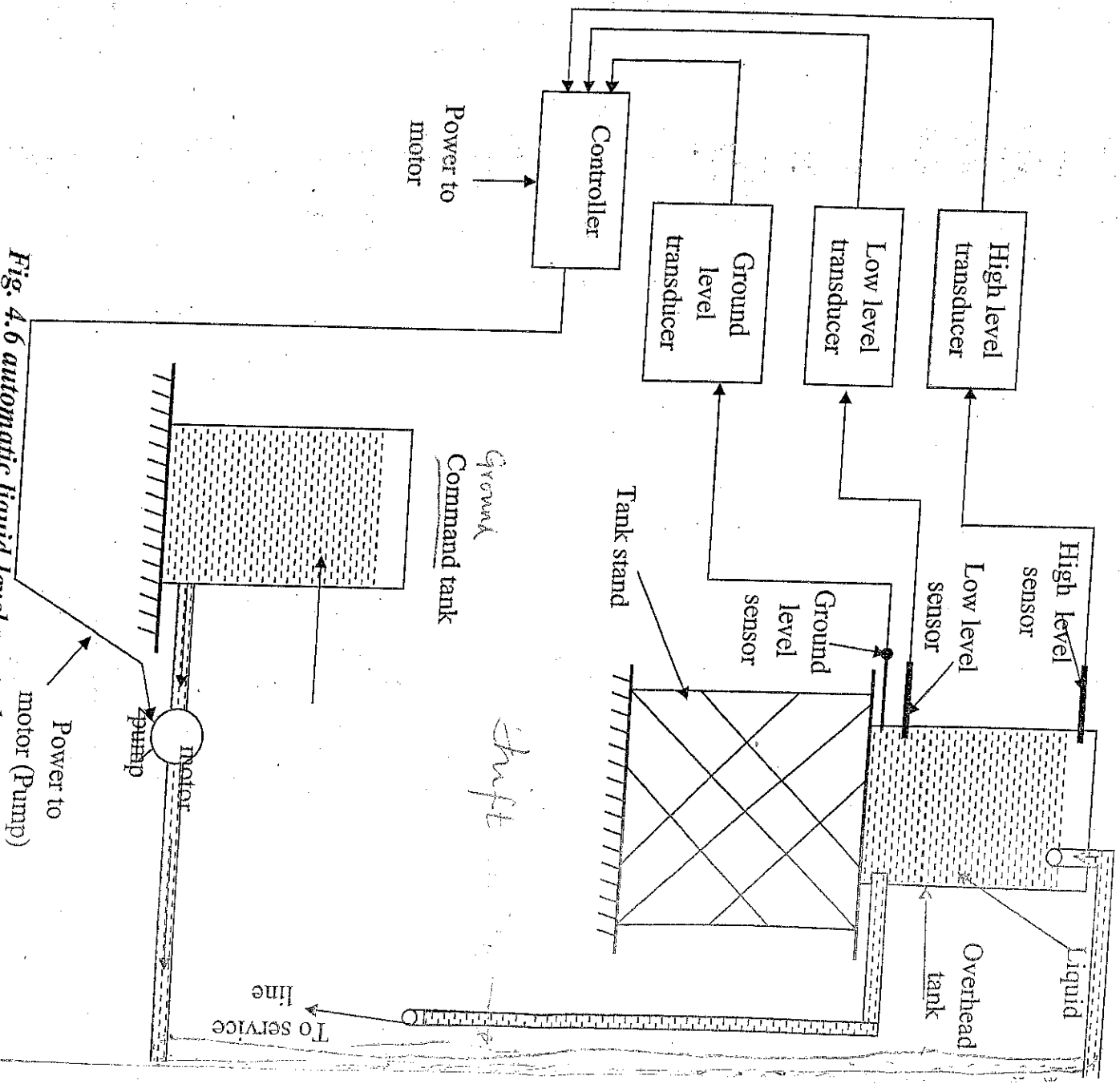


Fig. 4.6 automatic liquid-level control process

An automatic liquid level control process is shown in fig. 4.6. The intention is to ensure that the liquid level in the overhead tank does not go below the low level or

above the high sensors so as to ensure that there is always liquid in the overhead tank to service the liquid line.

4.7

Should the level of the liquid in the overhead tank tend to go below the low level sensor, a signal is released and applied to the controller through the low level transducer. The controller compares the signal with the setpoint and closes the power circuit to the overhead tank until the liquid level reaches the high level sensor. At this instant, a signal is sent through the high level transducer to the controller. The controller compares this signal with the setpoint and senses that the liquid level in the overhead tank has reached the highest level. Consequently, it opens the power circuit to the pump, thereby stopping pumping of liquid to the overhead tank. By this process the liquid level in the overhead tank is automatically controlled.



FIG. 4.7

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4.7 Automatic pressure control process

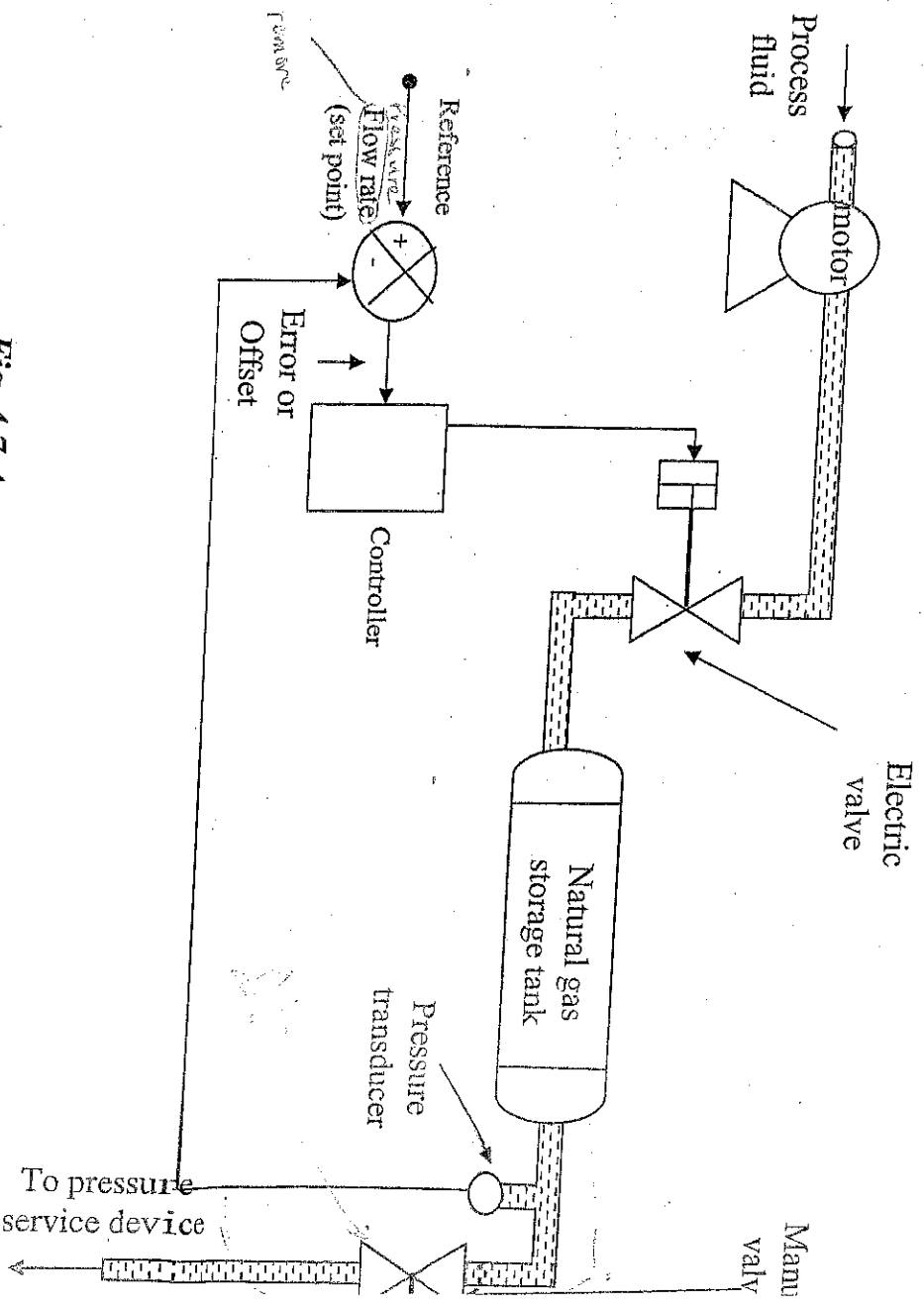


Fig. 4.7 Automatic pressure control system

Fig. 4.7 shows the diagram of the principle of operation of automatic pressure control process. This system finds application in natural gas supply facilities, when for example cooking gas cylinders are refilled. The motor pumps the process fluid to the storage tank through the electric valve which determines the pressure of the fluid stored in the tank. The reference pressure determined by the set point is applied through the controller to the electric valve.

The output pressure to the service line is converted by the pressure transducer to its equivalent electrical signal and fed back to the comparator. The output pressure compares with input or reference pressure and if there is a deviation or offset, it is

applied to the controller which adjusts the electric valve in such a manner that the pressure of the fluid pumped into the tank reverts to the desired value. The manual valve is opened to allow the fluid to service outlets at the desired pressure after which it is closed.

4.8 Automatic flow-rate control process

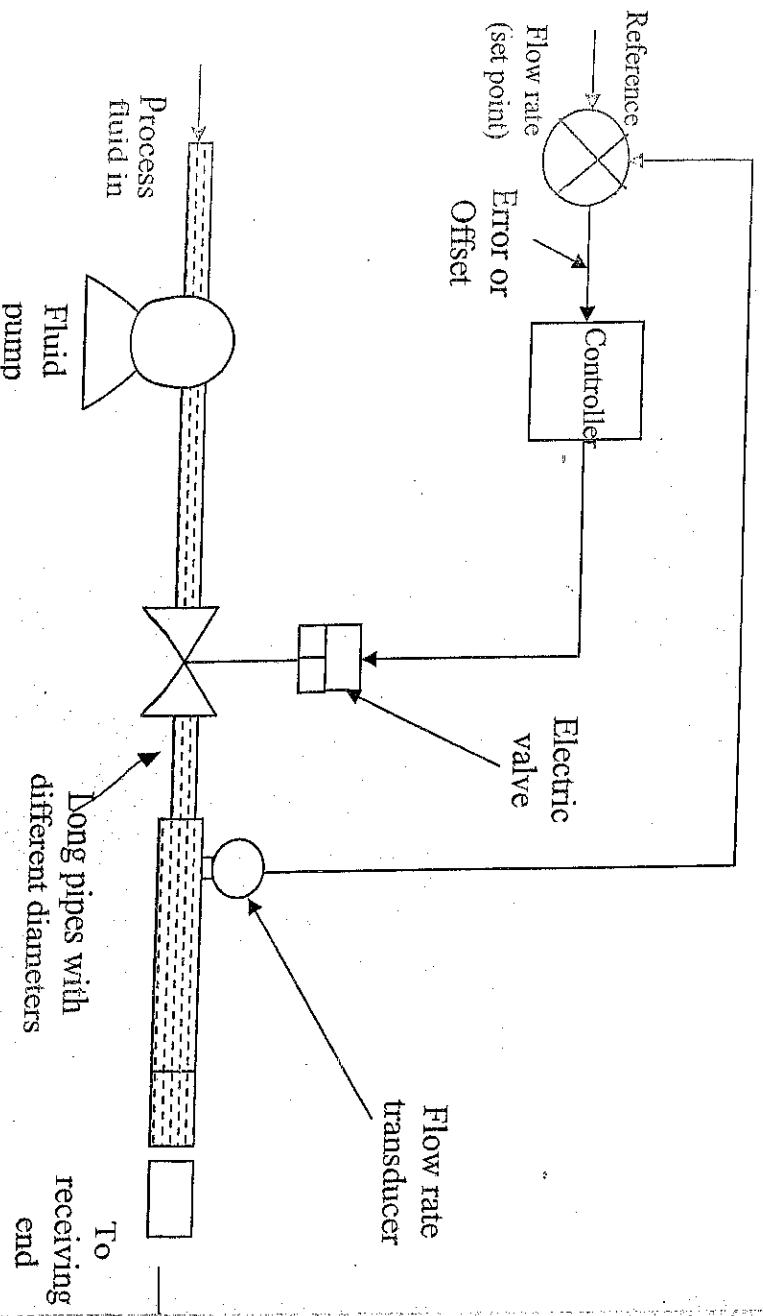


Fig. 4.8 automatic flow-rate control process

Fig. 4.8 illustrate the principle of operation of automatic flow-rate control process. This system finds application in pumping crude oil from its source to refineries several kilometers away from the source. In the figure, the desired rate of flow is set and applied to the “+” terminal of the comparator and the output of the comparator feeds the controller which commands the electric valve to adjust to the required

pressure so as to give the desired flow rate as the motor pumps the process fluid through the pipe line to the receiving end. A flow-rate transducer attached to the serving pipeline converts the rate of flow to its equivalent electrical signal and applied to the " - " input terminal of the comparator. This feedback signal compares with the reference input flow-rate and if there is a difference, an error signal is generated and fed to the controller which commands the electric valve to adjust its pressure in a manner that will restore the flow rate to the desired value. By this process, the flow-rate is automatically maintained constant.

4.9 Automatic power plant control process

Automatic power plant control process is shown in fig. 4.9. From the diagram it can be seen that the process variables to be controlled are the actual generation, turbine speed, boiler pressure, temperature, oxygen, etc. The computer performs a dual role as comparator and controller. The desired values of the process variable are programmed into the computer. similarly, the actual values of temperature, pressure, oxygen, turbine speed, and generation are fed back to the computer through their respective transducers. The computer compares the feedback values with the desired programmed values and takes decisions on how to correct deviations. The corrective signals are used to adjust the electric values and speed governor so as to restore the actual ^{values} ~~vel~~ues to the set point. These sequence of actions automatically control the power plant.

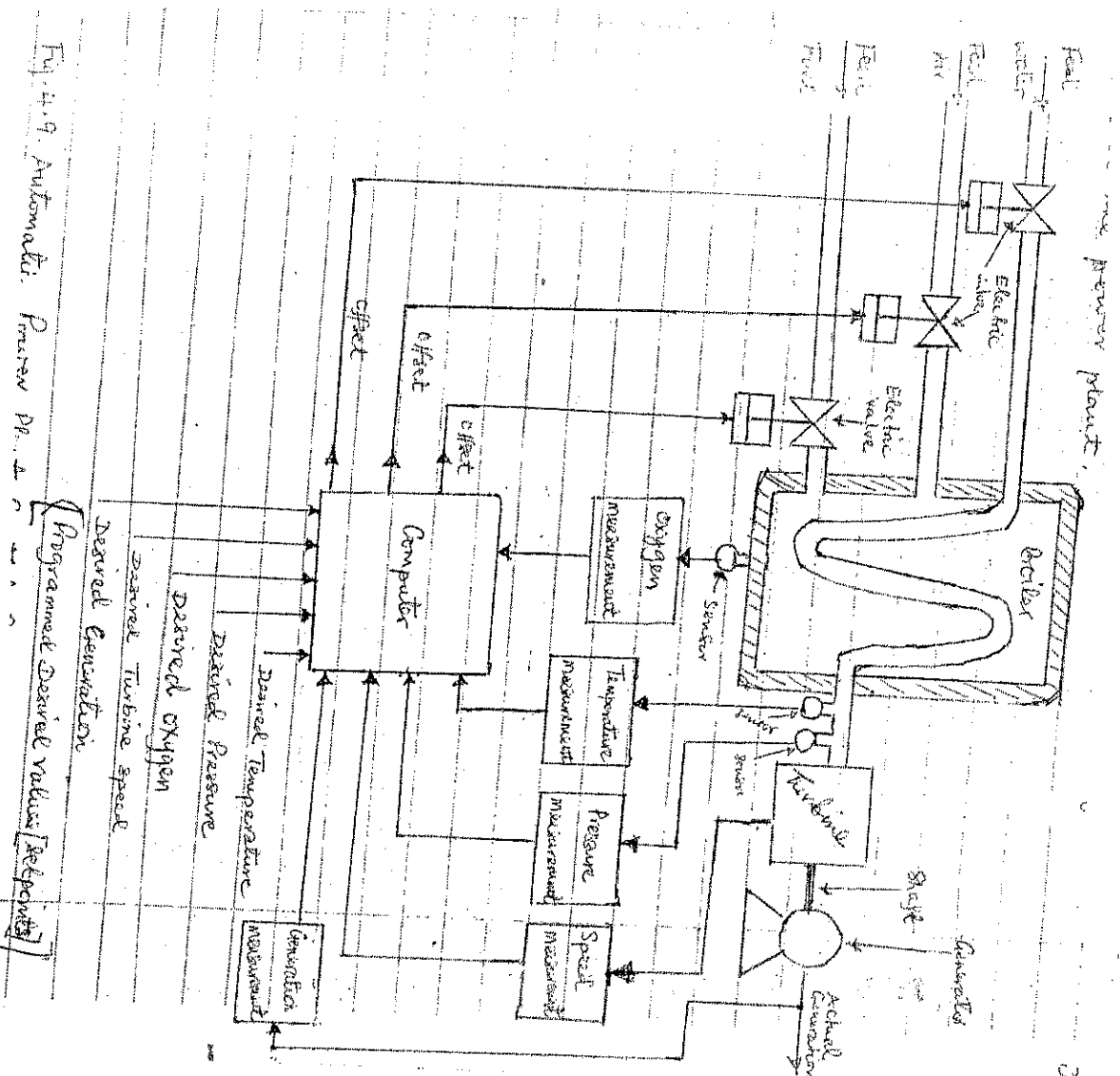


Fig. 4.9. Automatic power plant control process.

5.0 Analogue computer

The word analogue derives from 'analogy', and analogue computers were so-called because they were used to build module that created an analogy of the physical phenomena. Analogue computer is a form of computer that uses the continuously changeable aspects of physical phenomena such as temperature, pressure, speed, flow-rate, density, composition, etc which are converted to their electrical equivalent

and used to model the problem being solved. Physical events are represented by mathematical operations and the analogue computers perform continuous manipulations equivalent to the mathematical operations required. The result of these manipulations represent the physical phenomena, whose magnitude and behavior represents the solution to the problem.

There exists on the analogue computer a complete analogue between the physical quantities, events, and the mathematical numbers and manipulations. If many events are taking place at the same time in the physical world, they will also take place at the same time, or in parallel, on the analogue computer. An analogue device will consequently, arrive at a result in a shorter time than a digital machine which must perform all its operations serially or after encoding. Analogue computers are characterized by continuous operations performed in parallel, as opposed to digital computers which are discrete and serial devices.

Analogue computers are widely used in scientific and industrial applications and for experimental analysis of problems. During experimental analysis the analogue computers ^{allows} permits physical systems to be simulated and run at a faster rate than real time which could, in some instances, be hours, weeks, months or longer.

Analogue computers typically have front panel with numerous jacks (single-contact sockets) that permit patch cords (flexible wires with plugs at both ends) to create the interconnections which define the problem setup.

5.1 Basic analogue computer components

Basic analogue computer components include;

1. Operational amplifier (Op-Amp)
2. Resistors,
3. Potentiometer (attenuators),

so-called
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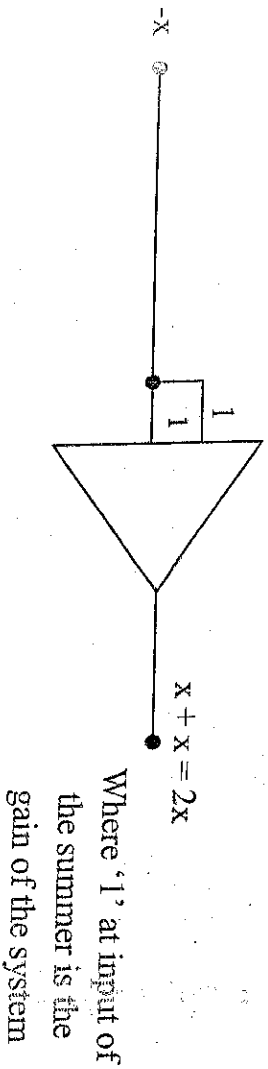
4. Capacitors,
5. Diodes,
6. Function generator,
7. Summing networks,
8. Integrating networks,
9. Differentiating networks,
10. Power supplies.
11. Various meters and display devices
12. A patch bag.

5.2 Mathematical operations of analogue computers

Analogue computers are powerful computing devices consisting of a collection of computing elements, each of which has some inputs and outputs and performs a specific operations such as addition, subtraction, integration, multiplication, etc. These elements can then be interconnected freely to form a model, an analogy of the physical problem that is to be solved. The various computing elements can be used to simulate variety of mathematical model of physical devices and synthesis.

5.3 Symbols and mathematical operations of analogue computers

5.3.1 Addition (summation)



5.3.2

-x

-x

-y

-x

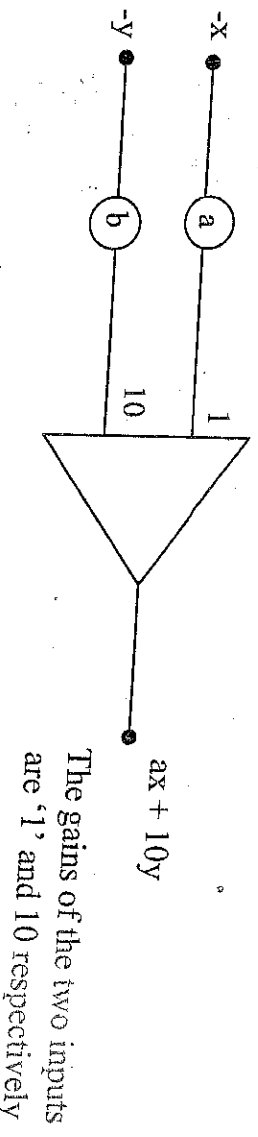
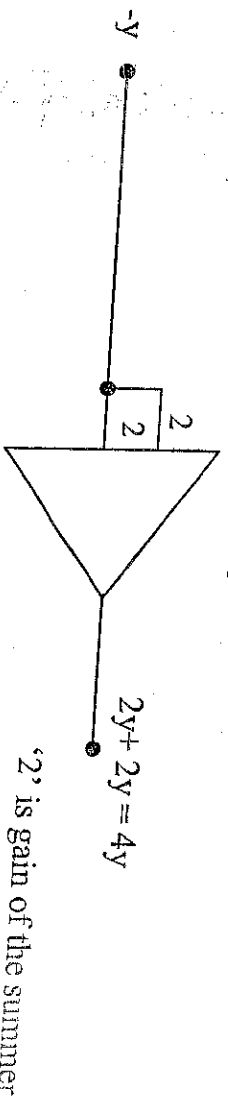
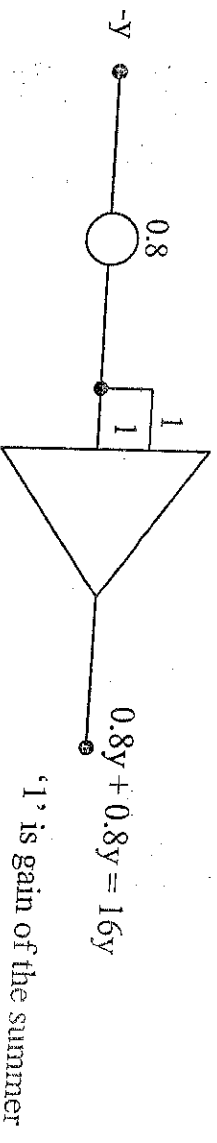
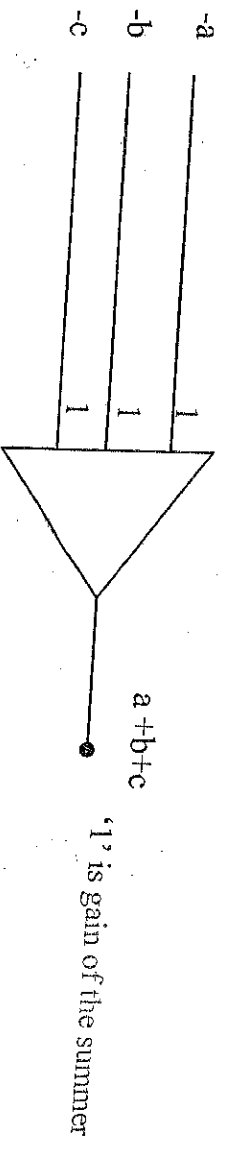
-y

-y

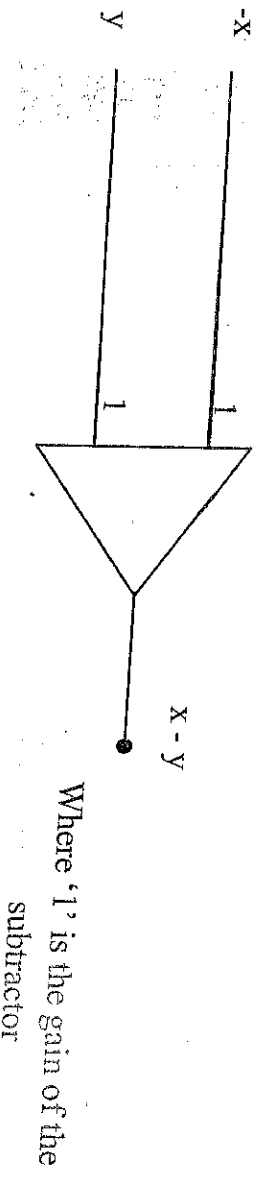
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-b

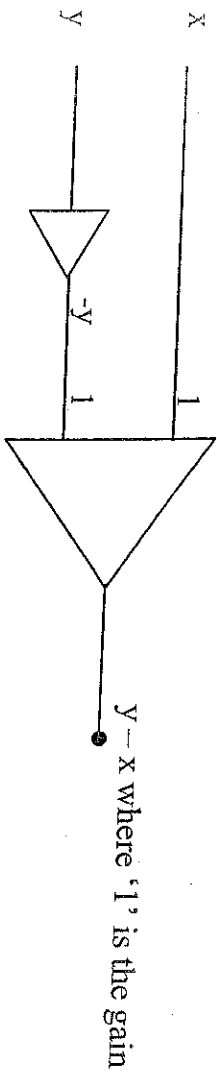
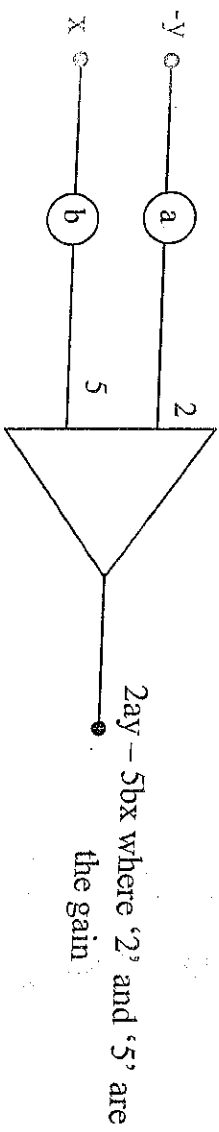
-a



5.3.2 Subtraction

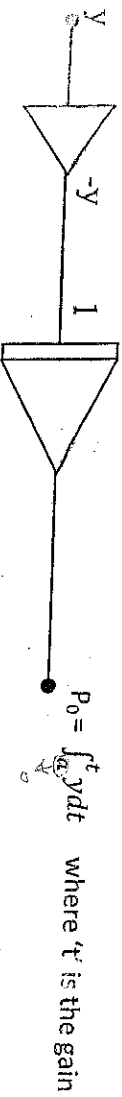
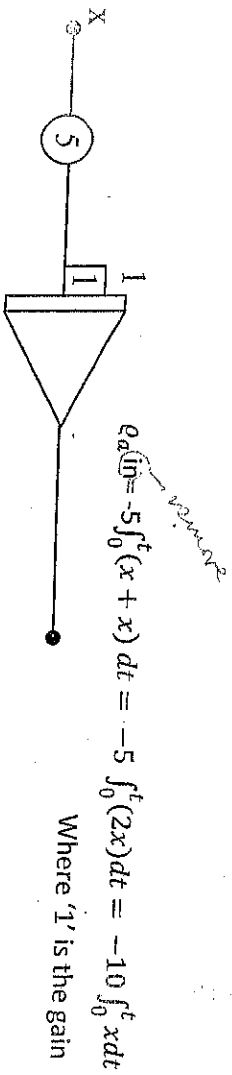
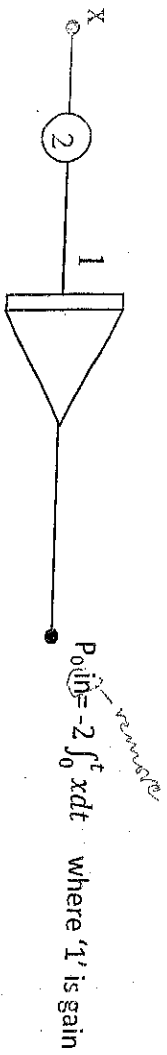
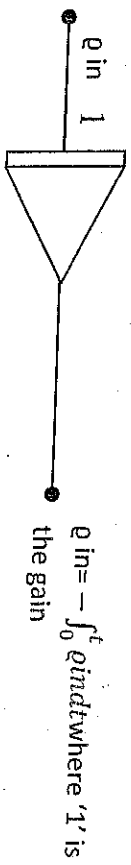


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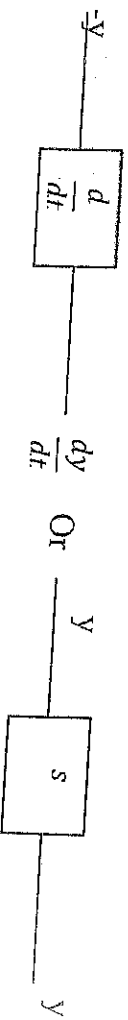


5.3.3 Integration

Integration symbol is

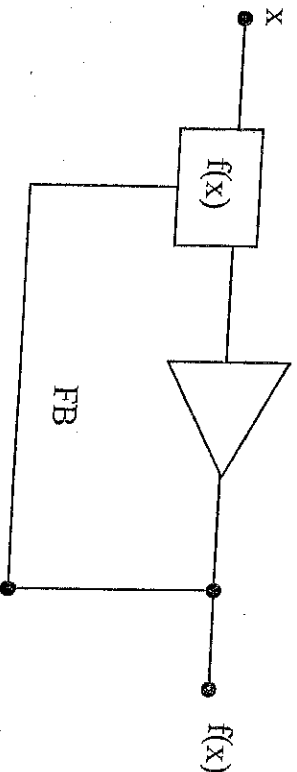


5.3.4 Differentiation:



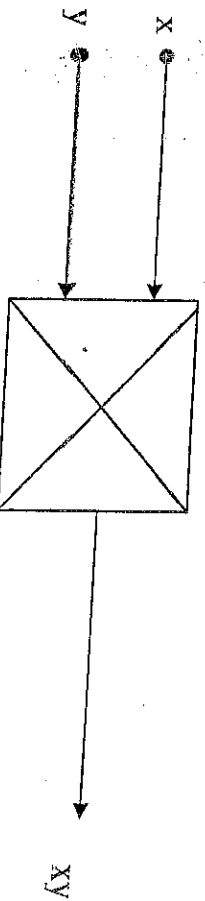
5.3.5 Function generation

The generation of arbitrary constants as well as analytical functions by means of function generations normally is required to obtain variable coefficients or parameters of a system, or to generate an input forcing function. Common types of functions are x^2 , x^4 , $\log x$, $\frac{1}{2} \log x$ ($=\log \sqrt{x}$), $\sin \frac{x}{\cos x}$, etc. The programming symbol for a fixed function generator is



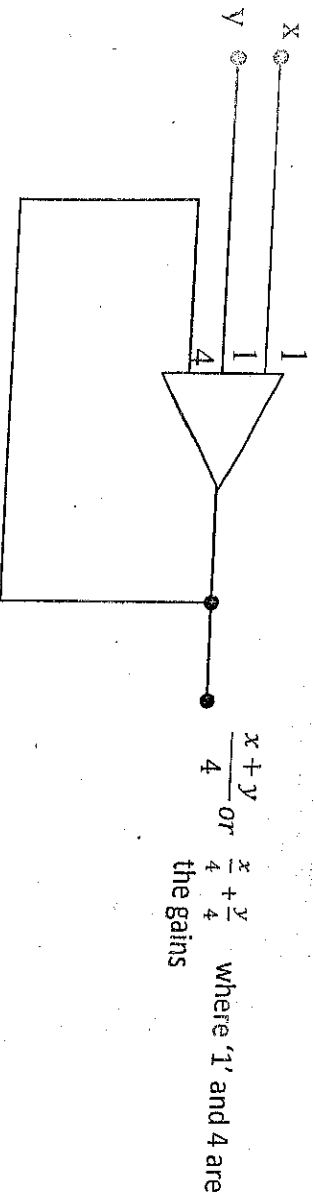
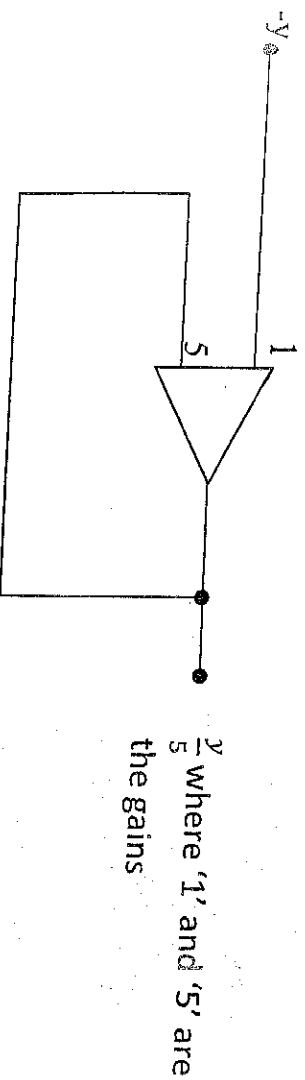
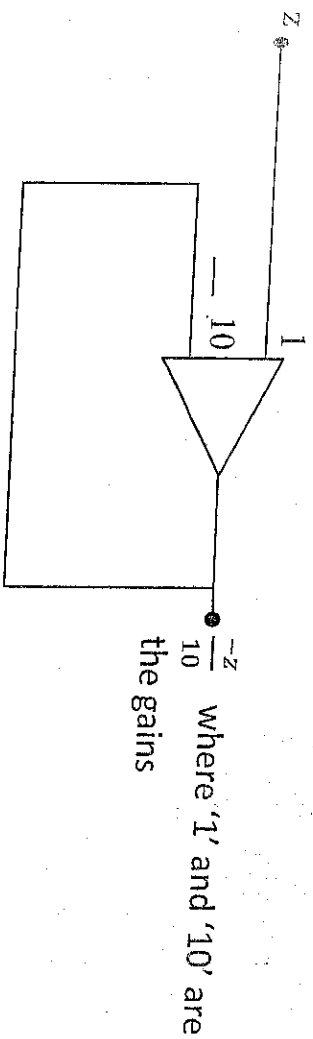
5.3.6 Multiplication

The programming symbol for multiplication is



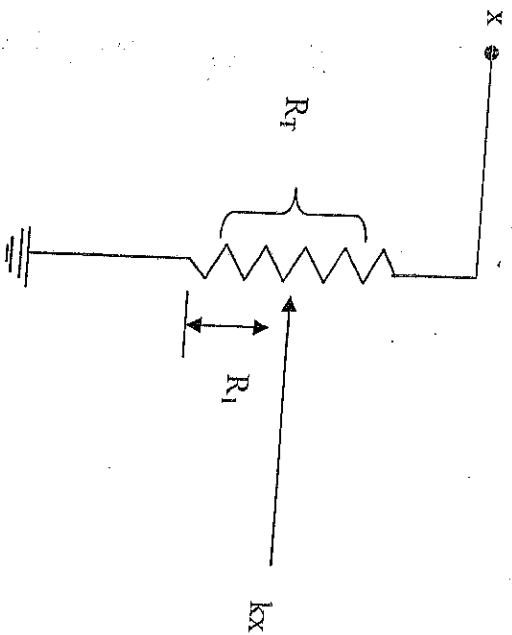
5.3.7 Division

Division operation by analogue computer is realized as follows;



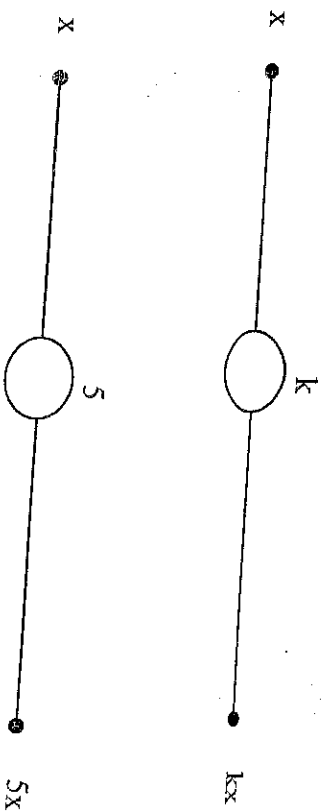
5.3.8 Multiplication by a constant

Multiplication by a constant which is less than unity accomplished by a potentiometer, also called an attenuator. This device is shown below simply it is a fixed resistor with a movable wiper arm.

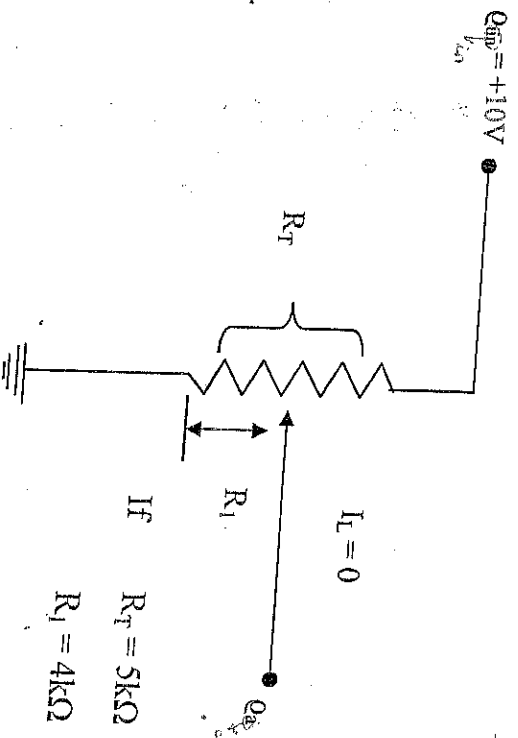


Where $k = \frac{R_1}{R_T}$

The programming for multiplication by a constant is as shown below



Example



Then, $Q_0 = 10 \left(\frac{R_1}{R_T} = \frac{4}{5} \right)$

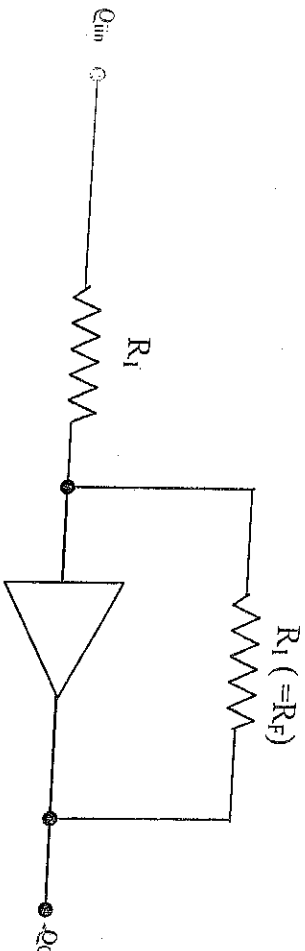
$$= 10 \times \frac{4}{5}$$

$$= 10 \times 0.8$$

$$= 8 \text{ volts}$$

If $R_T = 5k\Omega$
 $R_1 = 4k\Omega$

5.3.8 Phase inversion



The circuit above is used in analogue computer to accomplish phase inversion. Where $R_1 = R_F$ in magnitude, the amplifier output voltage has the same amplitude as the input voltage but is of opposite polarity. Thus the mathematical operation of inversion is performed. That is,

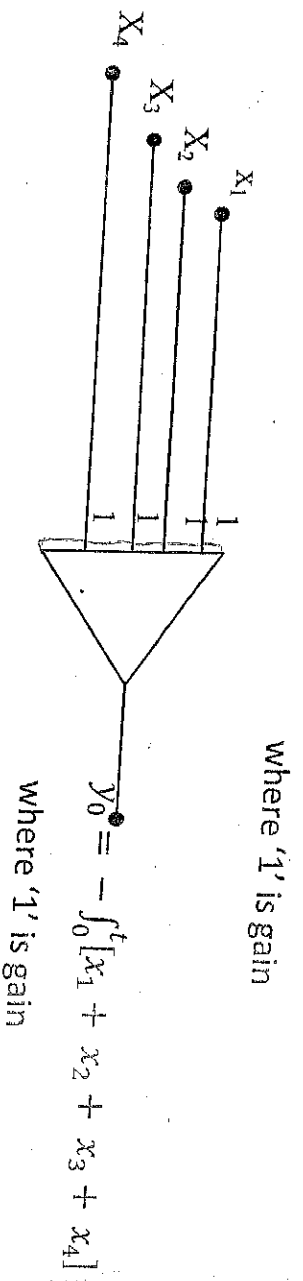
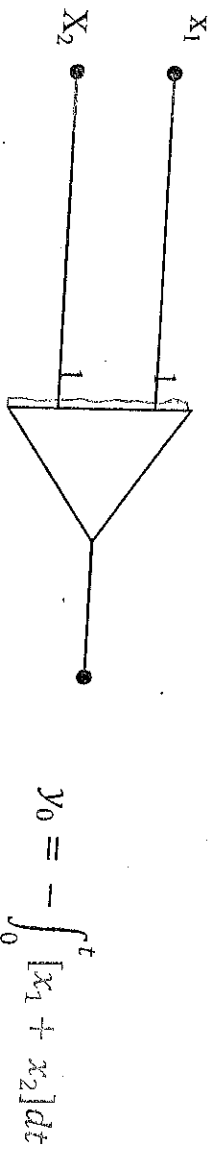
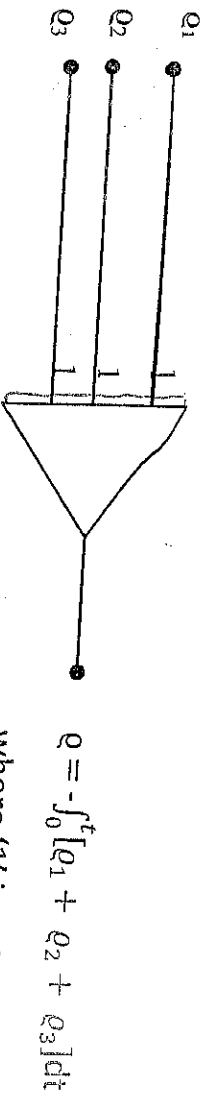
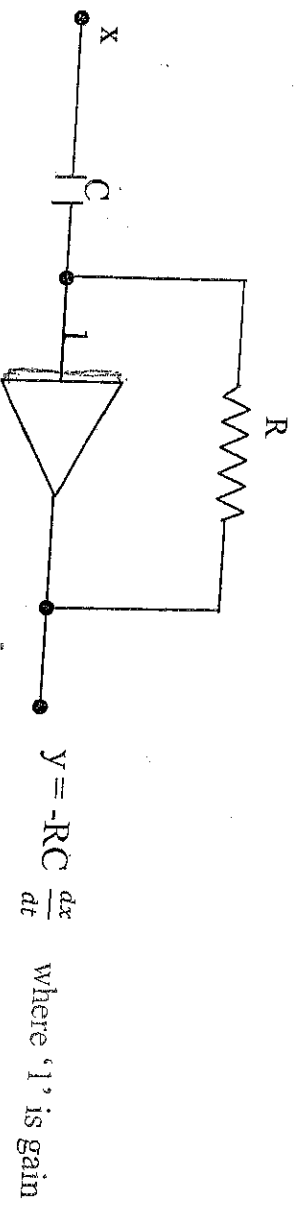
$$e_o = \frac{R_1}{R_1} e_{in} = -e_{in}$$

The programming symbol for inversion operation is



5.3.10 Summation integration

Summation integration operation is performed when multiple inputs are applied to the integrator, which gives an output that equals the integration of the sum of the various inputs as shown below



5.3.11 Differentiation circuit

5.4 Analogue computer flow diagrams to simulate equations

5.4.1 Linear equation

$$3x + 4y = 10 \quad (1)$$

Before one can draw the flow diagram the expression for x and y must be solved from the equation as follows;

$$3x = 10 - 4y$$

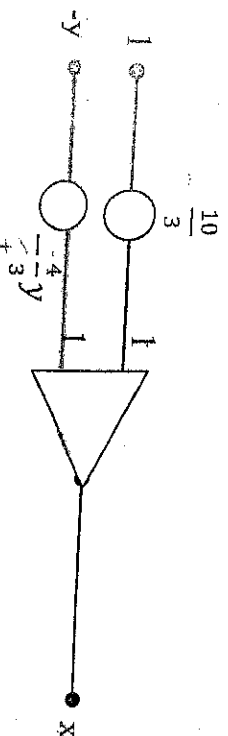
$$\therefore x = \frac{10}{3} - \frac{4}{3}y \quad (2)$$

Or

$$4y = -10 - 3x$$

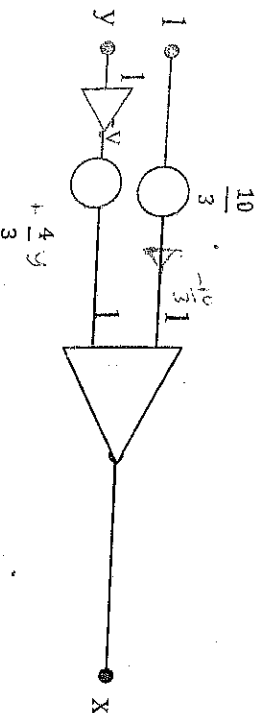
$$y = \frac{10}{4} - \frac{3}{4}x \quad (3)$$

The analogue computer flow diagram that simulates equation (2) and (3) which gives its solutions is as shown in fig. 5.4.1.

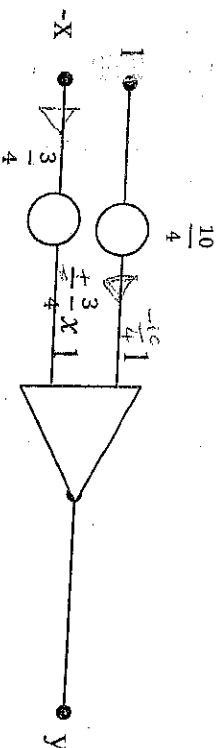


where '1' is gain

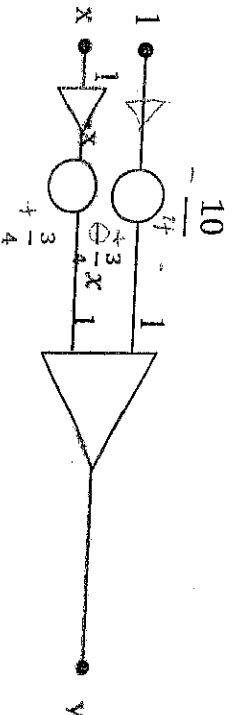
$$(for x = \frac{10}{3} - \frac{4}{3}y)$$



where '1' is gain



where '1' is gain



$$\left[\text{for } y = \frac{10}{4} - \frac{3}{4}x \right]$$

Fig. 5.4.1 Analogue computer flow diagram for solving equation $3x + 4y = 10$

Exercise

Simulate the analogue computer flow diagram for solving the equation $5x + 7y = 17$

5.4.6 Simultaneous equation

$$a_{11}X_1 + a_{12}X_2 = b_1 \quad - \quad - \quad - \quad - \quad - \quad (1)$$

$$a_{21}X_1 + a_{22}X_2 = b_2 \quad - \quad - \quad - \quad - \quad - \quad (2)$$

The unknowns are solved from the above equation as follows;

$$a_{21}X_1 = b_1 - a_{12}X_2$$

$$\therefore X_1 = \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} X_2 \quad (3)$$

Similarly,

$$a_{22} X_2 = b_2 - a_{21} X_1$$

$$\therefore X_2 = \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} X_1 \quad (4)$$

The analogue computer flow diagram that simulates the simultaneous equation (1) and (2) which gives its solution is as shown in fig. 5.4.2

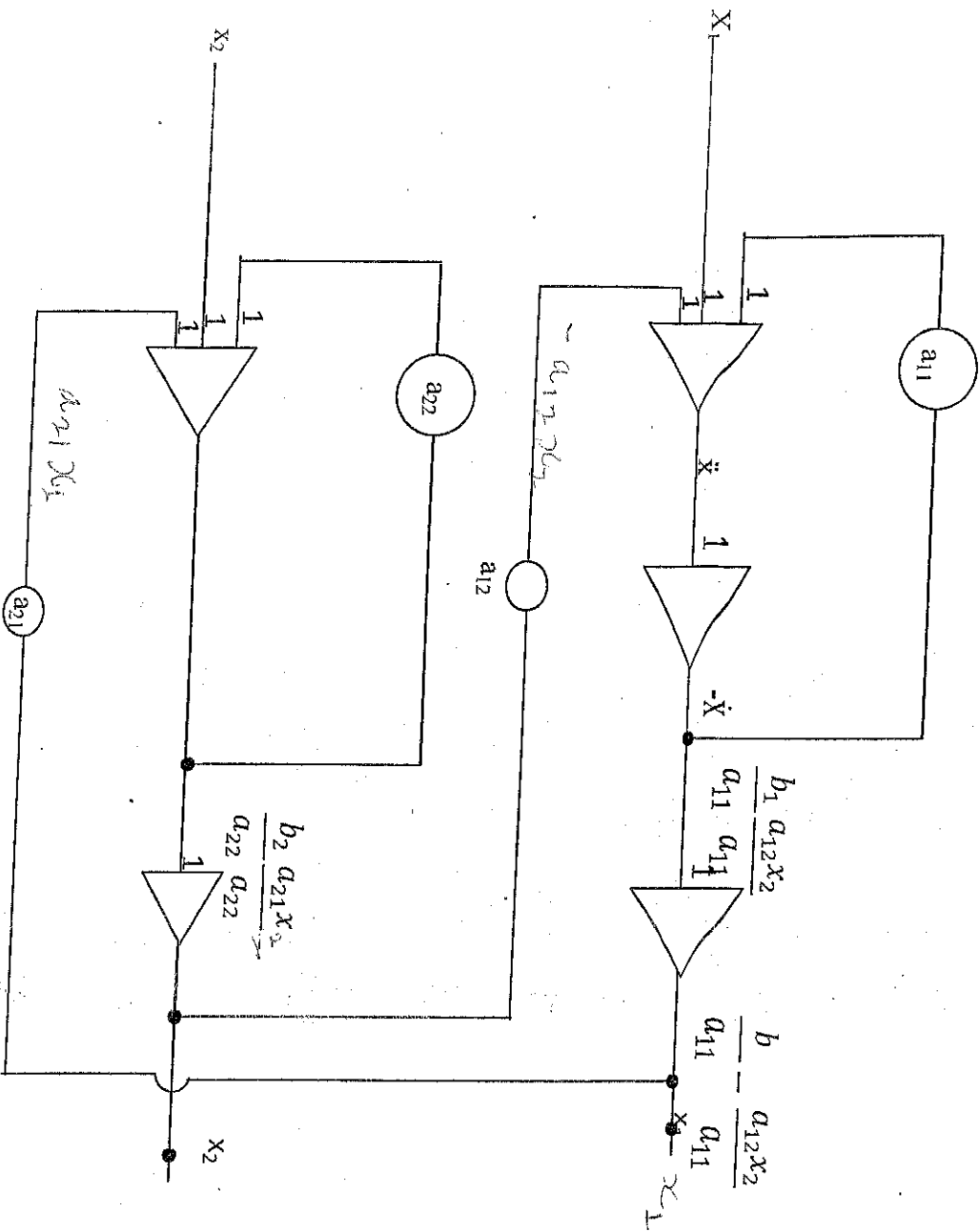


Fig. 5.4.2 *Analogue computer flow diagram for solving these simultaneous equations;*

$$a_{11}X_1 + a_{12}X_2 = b_1$$

$$a_{21}X_1 + a_{22}X_2 = b_2$$

Exercise

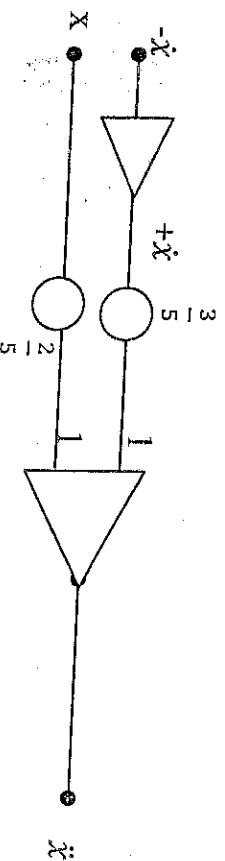
Simulate the analogue computer flow diagram for solving the following simultaneous equations.

$$2x + 3y = 5$$

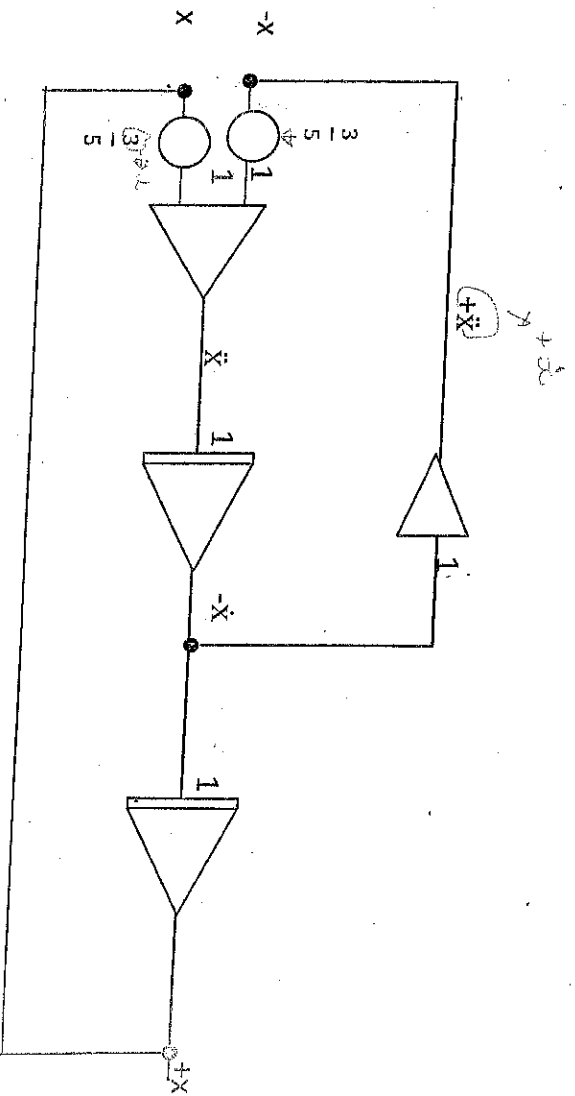
$$5x + 4y = 13$$



where '1' is the gain



where '1' is the gain



5.4.3 Differentiated ^{ied} equations

Many systems in nature are described by differential equations. These equations can be solved by interconnections of the blocks of analogue computer. This block diagram solver is called simulated analogue computer (SAC). The simulation provides the solution of these differentiated equations thereby giving the information a model contains.

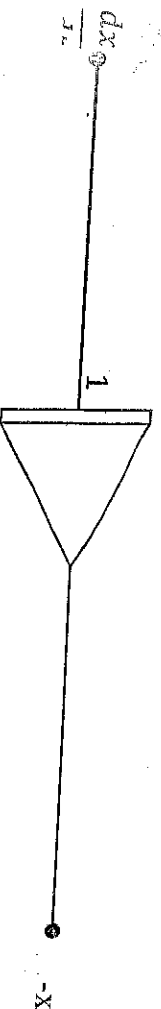
First order equation

A first order differentiated equation is as given below,

$$\frac{dx}{dt} = +kx = 0 \text{ (where } k \text{ is constant), hence,}$$

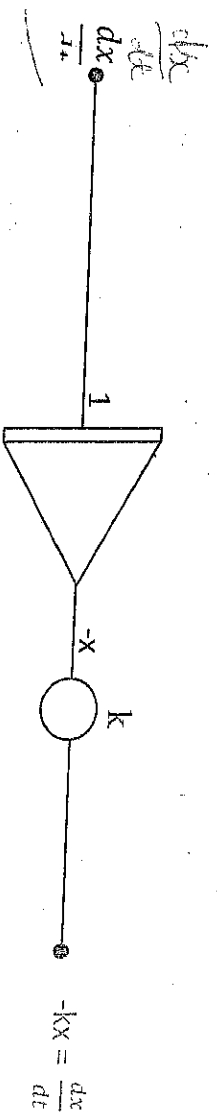
$$\frac{dx}{dt} = -kx \quad (1)$$

The two variables in these equation are x and $\frac{dx}{dt}$. The analogue computer can represent the relation between these variables with the integrator as follows,



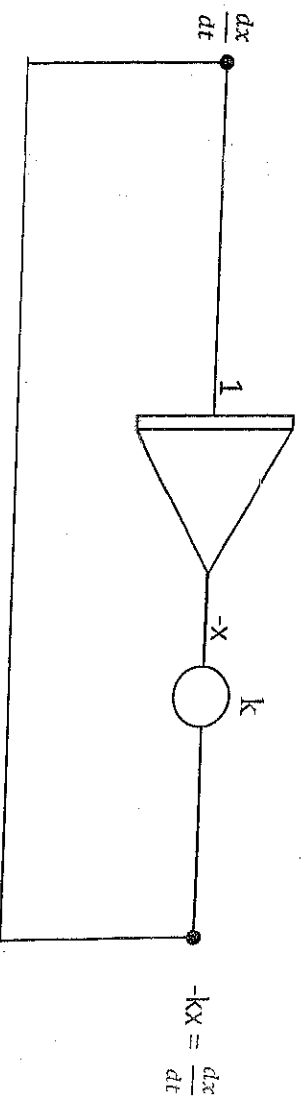
where '1' is the gain

It shows that if we feed $\frac{dx}{dt}$ into an integrator, the output will be $-x$. But where can we obtain $\frac{dx}{dt}$? From equation (1) we see that this is simply $-kx$. Since $-x$ is available at the integrator output, we can use a potentiometer to multiply it by k . Hence,



This produces $-kx$ at the output of the potentiometer

Since this is exactly what we needed at the integrator input, we can provide the desired input by connecting the potentiometer output to the integrator input as follows;



This final diagram is the analogue computer flow diagram that simulates the first order differentiated equation $\frac{dx}{dt} = -kx$ which gives the solution of the equation.

(b) Second order equation

A second order differential equation is given hereunder as

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = 0$$

$$\text{Or } \ddot{x} + b\dot{x} + cx = 0 \quad (2)$$

This equation may be solved on the analogue computer essentially the same manner as equation (1). In this case, since the equation is second order, we must integrate $\ddot{x}$ to produce $\dot{x}$, and integrate $\dot{x}$ to produce x as follows

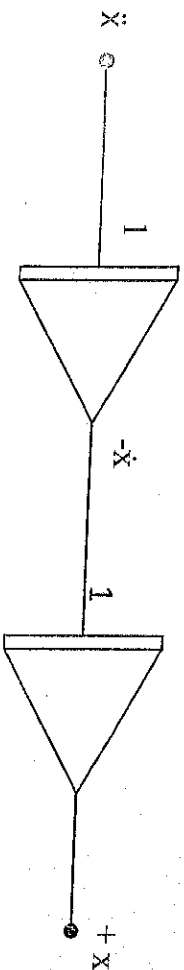


Fig. 1.0

How do we provide the necessary input $\ddot{x}$? To do this, we must solve equation (2) for $\ddot{x}$, hence,

$$a\ddot{x} = -(b\dot{x} + cx)$$

$$\ddot{x} = -\frac{1}{a}(b\dot{x} + cx) = -\frac{b}{a}\dot{x} - \frac{c}{a}x$$

this result says that we can obtain $\ddot{x}$ from $\dot{x}$ and x by means of a summing amplifier and two potentiometer as follows in fig 2 and Fig 3.

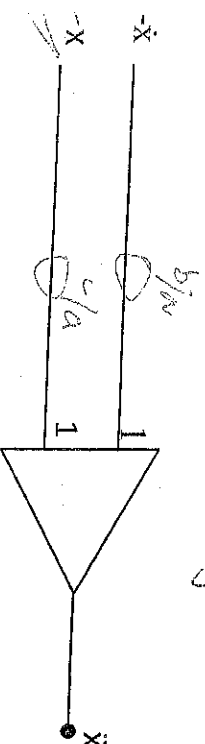


fig.2

To close the loop, we must combine fig. 1 and fig. 2 as shown below.

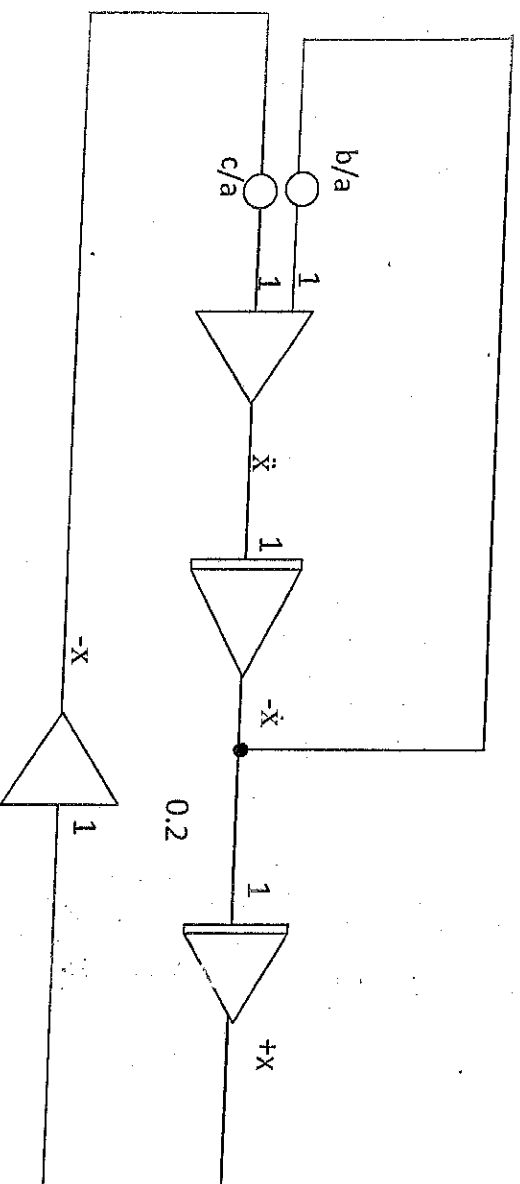


Fig. 3.0